Cart lab: change driving force

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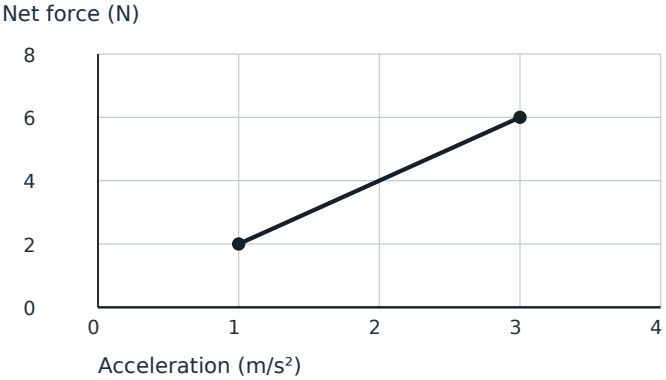
Learning target: Use controlled data to describe how net force affects acceleration.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 9.*
The straight-moving rover can have zero net force. The turning rover changes velocity direction and needs net force.

02. Worked example: explain the model
Plot F against a: (1, 2),(3, 6); slope 2 kg, the mass.

Model / diagram check



03. Read a proportional pattern
It doubles; at 8 N, a=4 m/s².

04. Keep mass constant
Driving weight and total moving mass change. Transfer mass from cart to hanger so total stays constant.

05. Graph the pattern
Line through (2, 1),(4, 2),(6, 3); ideally through origin. Axes force N anda m/s².

06. Review: connect an earlier idea *Review: Unit 1, Day 11.*
5 m/s²; the mass doubles at the same net force. Keep the net force fixed, including resistance.

07. Exit check
3 m/s² because force is 1.5 times as large at same mass.

08. STRETCH: test the model
3, 6, 9 N; total moving mass and relevant conditions fixed.

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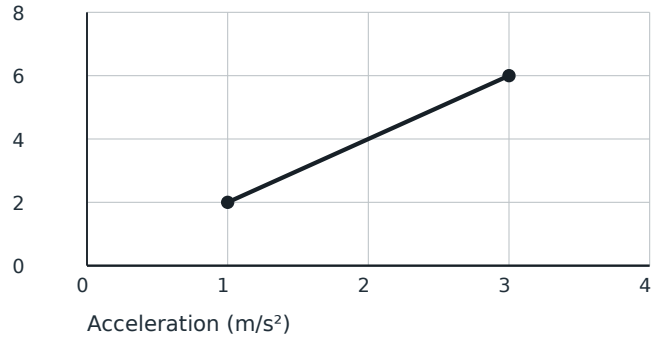
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01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 9.*
Tangent velocity direction changes, so acceleration and net force are inward and nonzero. Constant speed is insufficient for equilibrium.

02. Worked example: explain the model
Plot F against a: (1, 2),(3, 6); slope 2 kg, the mass.

Model / diagram check



03. Interpret a force graph
Slope 1(m/s²)/N=1/kg; mass 1 kg.

04. Design the controlled trial
Same parts remain in system; their location changes. Table m_h, F_{drive}, M, 3 a trials, mean. Keep release/track consistent; record friction effects.

05. Both variables change
aA=2 m/s²; aB=3 m/s²; ratio 1.5. Mass also doubles.

06. Review: connect an earlier idea *Review: Unit 1, Day 11.*
Net force 0.15 N; new a≈0.2143 m/s². Extra resistance reduces net force and acceleration if driving force is unchanged.

07. Exit check
Mass .50 kg; a=3 m/s². Resistance subtracts from driving force, so acceleration may not pass through origin on driving force plot.

08. STRETCH: test the model
F_{pull}=2 a +1, so resistance 1 N. Intercept at a=0 is 1 N; slope mass 2 kg.

Model / diagram check

